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10 CONCEPTUAL Q&A IOT & EDGE · PRACTICE WORKSHEET

CONCEPTUAL SERIES
ESP32

Name: ______________________ Cohort/Batch: ______________________ Date: ______________ Score: ____

ESP32 PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A IoT & Edge

TOTAL: 10 QUESTIONS

Q1 What is ESP32 and what problem in IoT & Edge does it solve? *Threat Model*

Q2 What are the core components or concepts in ESP32? *Architecture*

Q3 How does ESP32 compare to its main alternatives? *Configuration*

Q4 How do you get started with ESP32 for a new project? *Hardening*

Q5 What configuration options most affect ESP32's behavior in production? *SSO / IdP*

Q6 How does ESP32 handle reliability and high availability? *Observability*

Q7 What observability does ESP32 provide out of the box? *Lifecycle*

Q8 What are common performance bottlenecks in ESP32? *Compliance*

Q9 What security best practices apply when running ESP32? *Testing*

Q10 What mistakes do teams commonly make when adopting ESP32? *Tuning*

ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 ESP32 is a tool or platform that
Mitigation

ESP32 is a tool or platform that automates or simplifies a specific concern in the IoT & Edge domain, reducing manual effort and operational complexity for engineering teams.

A2 ESP32 has key building blocks that work
Primitives

ESP32 has key building blocks that work together: a configuration layer defines intent, a runtime layer enforces it, and an API or CLI layer allows operators to manage and observe the system.

A3 ESP32 trades off simplicity against feature richness
Hardening

ESP32 trades off simplicity against feature richness differently than its alternatives. Choose ESP32 when its specific strengths performance, ecosystem, or ease of use align with your team's primary constraints.

A4 Install ESP32 via its package manager or
Gotchas

Install ESP32 via its package manager or binary release. Follow the quickstart to initialize a project, configure the minimal required settings, and run the default command to verify the setup works end-to-end.

A5 Production-critical settings typically include concurrency
Federation

Production-critical settings typically include concurrency limits, timeout values, retry policies, resource limits, and logging levels. Review the official production hardening guide before deploying.

A6 ESP32 provides HA through leader election, replication,
Observability

ESP32 provides HA through leader election, replication, or stateless horizontal scaling. Deploy at least two instances across availability zones and configure health checks for automatic failover.

A7 ESP32 exposes metrics (Prometheus endpoint or built-in
Automation

ESP32 exposes metrics (Prometheus endpoint or built-in dashboard), structured logs, and optionally distributed traces. Define SLOs on the key metrics and alert before users are affected.

A8 Performance bottlenecks typically stem from misconfigured
Governance

Performance bottlenecks typically stem from misconfigured connection pools, large payload sizes, insufficient worker threads, or missing caches. Profile with built-in diagnostics before optimizing.

A9 Run ESP32 with the principle of least
Assessment

Run ESP32 with the principle of least privilege, enable TLS for all communication, use a secrets manager for credentials, and keep the software patched to the latest stable release.

A10 Common pitfalls include skipping the documentation and
Optimization

Common pitfalls include skipping the documentation and learning the API by trial and error, using default configurations in production, lacking a runbook for operational issues, and not testing failover scenarios.